Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

$x + ay = a^2 + 1$ is a tangent to the circle $x^2 + y^2 = 10$ for

(A) any values of a

(B) only one value of a

(C) two values of a

(D) no value of a

Q.2

If the circles $x^2 + y^2 + 2ax + cy + a = 0$ and $x^2 + y^2 - 3ax + dy - 1 = 0$ intersect in two distinct points P and Q then the line $5x + by - a = 0$ passes through P and Q for

(A) infinitely many values of a

(B) exactly two values of a

(C) exactly one value of a

(D) no value of a

Q.3

If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $\mathrm{P}\left(\mathrm{x}_1, \mathrm{y}_1\right), \mathrm{Q}\left(\mathrm{x}_2, \mathrm{y}_2\right), \mathrm{R}\left(\mathrm{x}_3, \mathrm{y}_3\right), \mathrm{S}\left(\mathrm{x}_4, \mathrm{y}_4\right)$ then

(A) $x_1 + x_2 + x_3 + x_4 = 0$

(B) $y_1 + y_2 + y_3 + y_4 = 2$

(C) $x_1 x_2 x_3 x_4 = 2c^4$

(D) $y_1 y_2 y_3 y_4 = 2c^4$

Q.4

The angle of intersection of the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 2x + 2y$ is

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{4}$

@IITJEE_Advanced

Q.5

Let C_1 and C_2 be two circles of radius r_1 and r_2 respectively $r_1 > r_2$ touching both the axes. If the two circles are orthogonal, then $\frac{r_1}{r_2}$ is equal to

(A) 2

(B) $2+\sqrt{3}$

(C) $3+\sqrt{2}$

(D) 3

Q.6

If a circle passes through the point (a, b) and cuts the circle $x^2+y^2=4$ orthogonally, then the locus of its centre is

(A) $2ax-2by-\left(a^2+b^2+4\right)=0$

(B) $2ax+2by-\left(a^2+b^2+4\right)=0$

(C) $2ax-2by+\left(a^2+b^2+4\right)=0$

(D) $2ax+2by+\left(a^2+b^2+4\right)=0$

Q.7

The circle $x^2+y^2=4x+8y+5$ intersects the line $3x-4y=m$ at two distinct points if

(A) $-35<\mathrm{m}<15$

(B) $15<\mathrm{m}<65$

(C) $35<\mathrm{m}<85$

(D) $-85<\mathrm{m}<-35$

Q.8

The equation of a circle which passes through the point $(2, 0)$ and whose centre is the limit of the point of intersection of the lines $3x+5y=1$ and $(2+c)x+5c^2y=1$ as c tends to 1, is

(A) $25\left(\mathrm{x}^2+\mathrm{y}^2\right)+20\mathrm{x}+2\mathrm{y}-60=0$

(B) $25\left(x^2+y^2\right)-20x+2y+60=0$

(C) $25\left(\mathrm{x}^2+\mathrm{y}^2\right)-20\mathrm{x}+2\mathrm{y}-60=0$

(D) None of these

Q.9

@IITJEE_Advanced

If the pair of lines $ax^2 + 2(a+b)xy + by^2 = 0$ lie along diameters of a circle and divide the circle into four sectors such that the area of one of the sectors is thrice the area of another sector then

(A) $3a^2 - 10ab + 3b^2 = 0$

(B) $3a^2 - 2ab + 3b^2 = 0$

(C) $3a^2 + 10ab + 3b^2 = 0$

(D) $3a^2 + 2ab + 3b^2 = 0$

Q.10

The line $y = mx + c$ intersects the circle $x^2 + y^2 = r^2$ at the two real distinct points if

(A) $-r\sqrt{1+m^2} < c < r\sqrt{1+m^2}$

(B) $-\mathrm{r} < \mathrm{c} < \mathrm{r}$

(C) $-r\sqrt{1-m^2} < c < r\sqrt{1+m^2}$

(D) None of these

Q.11

A line is drawn through a fixed point $P(\alpha, \beta)$ to cut the circle $x^2 + y^2 = a^2$ at A and B, then PA.PB is equal to

(A) $\alpha^2 + \beta^2$

(B) $\alpha^2 + \beta^2 - a^2$

(C) a^2

(D) $\alpha^2 + \beta^2 + a^2$

Q.12

The length of the chord intercepted by the circle $x^2 + y^2 = r^2$ on the line $\frac{x}{a} + \frac{y}{b} = 1$ is

(A) $\sqrt{\frac{r^2(a^2+b^2)-a^2b^2}{a^2+b^2}}$

(B) $2\sqrt{\frac{r^2(a^2+b^2)-a^2b^2}{a^2+b^2}}$

(C) $2\frac{\sqrt{r^2(a^2+b^2)-a^2b^2}}{a^2+b^2}$

(D) None of these

Q.13

If OA and OB are the tangents from the origin to

@IITJEE_Advanced

the circle $x^2+y^2+2gx+2fy+c=0$ and C is the centre of the circle, the area of the quadrilateral OACB is

(A) $\frac{1}{2}\sqrt{c\left(g^2+f^2-c\right)}$

(B) $\sqrt{c\left(g^2+f^2-c\right)}$

(C) $c\sqrt{g^2+f^2-c}$

(D) $\frac{\sqrt{g^2+f^2-c}}{c}$

Q.14

If from any point P, tangents $\mathrm{PT}, \mathrm{PT}'$ are drawn to two given circles with centres A and B respectively; and if PN is the perpendicular from P on their radical axis, then $\mathrm{PT}^2-\mathrm{PT}'^2=$

(A) PN.AB

(B) 2PN.AB

(C) 4PN. AB

(D) None of these

Q.15

Consider a family of circles which are passing through the point $(-1,1)$ and are tangent to x -axis. If (h,k) are the coordinates of the centre of the circles, then the set of values of k is given by the interval

(A) $-\frac{1}{2} \leq k \leq \frac{1}{2}$

(B) $k \leq \frac{1}{2}$

(C) $0<k<\frac{1}{2}$

(D) $k \geq \frac{1}{2}$

Q.16

The locus of the centres of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 5x + 4y - 2 = 0$ orthogonally is -

(A) $9x+10y-7=0$

(B) $x-y+2=0$

(C) $9x-10y-11=0$

(D) $9x+10y+7=0$

Q.17

If the line $x+y=1$ is a tangent to a circle with centre $(2,3)$, then its equation is

@IITJEE_Advanced

(A) $x^2+y^2+2x+2y+5=0$

(B) $x^2+y^2-4x-6y+5=0$

(C) $x^2+y^2-x-y+3=0$

(D) $x^2+y^2+5x+2y=0$

Q.18

If the equation of the common tangent at the point $(1,-1)$ to the two circles, each of radius 13 , is $12x+5y-7=0$ then the centres of the two circles are

(A) $(13,4),(-11,6)$

(B) $(13,4),(-11,-6)$

(C) $(13,-4),(-11,-6)$

(D) $(-13,4),(-11,-6)$

Q.19

What is the equation to circle which touches both the axes and has centre on the line $x+y=4$?

(A) $x^2+y^2-4x+4y+4=0$

(B) $x^2+y^2-4x-4y+4=0$

(C) $x^2+y^2+4x-4y-4=0$

(D) $x^2+y^2+4x+4y-4=0$

Q.20

The lengths of the tangent drawn from any point on the
circle $15x^2+15y^2-48x+64y=0$ to the two circles $5x^2+5y^2-24x+32y+75=0$ and $5x^2+5y^2-48x+64y+300=0$
are in the ratio of

(A) $1:2$

(B) $2:3$

(C) $3:4$

(D) None of these

Answers & Solutions

Q.1 Answer:

two values of a

Solution:

@IITJEE_Advanced

If $x+ay=a^2+1$ touches the circle, length of the perpendicular of the centre $(0,0)$ of the circle from the line is $\sqrt{10}$, the radius of the circle. $\Rightarrow \left|\frac{0+0.9-\left(a^2+1\right)}{\sqrt{1+a^2}}\right|=\sqrt{10}$

$\Rightarrow \quad \sqrt{1+a^2}=\sqrt{10} \Rightarrow a^2=9 \Rightarrow a=\pm 3$

Q.2 Answer:

no value of a

Solution:

The equation of the common chord is $5ax+(c-d)y+a+1=0$. This should be the same as $5x+by-a=0$. Hence $\frac{5a}{5}=\frac{c-d}{b}=\frac{a+1}{-a}$ which gives the quadratic $a^2+a+1=0$ which has no real roots.

Q.3 Answer:

$x_1+x_2+x_3+x_4=0$

Solution:

(a) $\left(\mathrm{x_i}, \mathrm{y_i}\right), \mathrm{i}=1,2,3,4$ lies on

$xy=c^2 \Rightarrow y_i=\frac{c^2}{x_i}$

Now the point $\left(\mathrm{x_i}, \mathrm{y_i}\right)$ lies on

$x^2+y^2=a^2 \Rightarrow x_i^2+\frac{c^4}{x_i}=a^2$

$\Rightarrow x_i^4-a^2 x_i^2+c^4=0$

Its roots are $x_1, x_2, x_3, x_4 \therefore$

$\mathrm{x_1}+\mathrm{x_2}+\mathrm{x_3}+\mathrm{x_4}=0$

$x_1x_2+x_1x_3+x_1x_4+x_2x_3+x_2x_4+x_3x_4=a^2$

$\mathrm{x_1x_2x_3}+\mathrm{x_1x_2x_4}+\mathrm{x_1x_3x_4}+\mathrm{x_2x_3x_4}=0$

$\mathrm{x_1x_2x_3x_4}=\mathrm{c}^4$ Clearly (c) is not correct

Now $\mathrm{y_1y_2y_3y_4}=\frac{\mathrm{c}^2}{\mathrm{x_1}} \cdot \frac{\mathrm{c}^2}{\mathrm{x_2}} \cdot \frac{\mathrm{c}^2}{\mathrm{x_3}} \cdot \frac{\mathrm{c}^2}{\mathrm{x_4}}=\mathrm{c}^4$

and $y_1+y_2+y_3+y_4=\frac{c^2\left(\sum x_1x_2x_3\right)}{x_1x_2x_3x_4}=0$

Q.4 Answer:

$\frac{\pi}{4}$

Solution:

Equations of the circles are $x^2+y^2=4$

and $x^2+y^2=2x+2y$

Centre of (1) is $\mathrm{C_1} \equiv (0,0)$; Radius of $(1)=\mathrm{r_1}=2$;

Centre of (2) is $\mathrm{C_2} \equiv (1,1)$; Radius of $(2)=\mathrm{r_2}=\sqrt{2}$

$\mathrm{d}=$ **distance between centres** $=\mathrm{C_1C_2}=\sqrt{1+1}=\sqrt{2}$

If θ is the angle of intersection of two circles, then

@IITJEE_Advanced

$\cos\theta = \frac{r_1^2+r_2^2-d^2}{2r_1r_2} = \frac{(2)^2+(\sqrt{2})^2-(\sqrt{2})^2}{2.2\cdot\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\therefore \theta = \frac{\pi}{4}$

Q.5 Answer:

$2+\sqrt{3}$

Solution:

Let the equation of a circle touching both the axes be

$x - r^2 + y - r^2 = r^2$

$x^2 + y^2 - 2rx - 2ry + r^2 = 0$

If the two given circles are orthogonal, then

$2\text{-}r_1\text{-}r_2 + 2\text{-}r_1\text{-}r_2 = r_1^2 + r_2^2$

$\Rightarrow r_1^2 - 4r_1r_2 + r_2^2 = 0$

$\Rightarrow \frac{r_1}{r_2} = 2+\sqrt{3}$

Q.6 Answer:

$2ax + 2by - \left(a^2 + b^2 + 4\right) = 0$

Solution:

Let the variable circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ **.....**

It passes through (a, b) $\therefore a^2 + b^2 + 2ga + 2fb + c = 0$ **. . . . . .**

(1) cuts $x^2 + y^2 = 4$ **orthogonally**

$\therefore 2(g \times 0 + f \times 0) = c - 4 \Rightarrow c = 4$

$\therefore$ **from (2)** $a^2 + b^2 + 2ga + 2fb + 4 = 0$

$\therefore$ **Locus of centre** $(-g, -f)$ **is** $a^2 + b^2 - 2ax - 2by + 4 = 0$

or $2ax + 2by = a^2 + b^2 + 4$

Q.7 Answer:

$-35 < \mathrm{m} < 15$

Solution:

Circle $x^2 + y^2 - 4x - 8y - 5 = 0$

Centre $= (2, 4)$**, Radius** $= \sqrt{4 + 16 + 5} = 5$

If circle is intersecting line $3x - 4y = m$ **at two distinct points.** $\Rightarrow \frac{|6-16-\mathrm{m}|}{5} < 5$

$\Rightarrow |10 + \mathrm{m}| < 5$

$\Rightarrow -25 < \mathrm{m} + 10 < 25$

$\Rightarrow -35 < \mathrm{m} < 15$

Q.8 Answer:

$25\left(\mathrm{x}^2 + \mathrm{y}^2\right) - 20\mathrm{x} + 2\mathrm{y} - 60 = 0$

Solution:

@IITJEE_Advanced

(c) Let $A \equiv (2,0)$

Given lines are $3x+5y=1$ and $(2+c)x+5c^2y=1$

Multiplying equation (1) by c^2 and subtracting

(2) form it, we get $\left(3c^2-c-2\right)x=c^2-1$ or

$x=\frac{c^2-1}{3c^2-c-2}$

Now,

$\lim_{c\to1}x=\lim_{c\to1}\frac{(c-1)(c+1)}{(c-1)(3c+2)}=\lim_{c\to1}\frac{c+1}{3c+2}=\frac{2}{5}$

$\therefore$ x coordinate of centre $=\frac{2}{5}$

From (1), when $x=\frac{2}{5}, y=-\frac{1}{25}$

Hence, the centre of the circle is $\left(\frac{2}{5},-\frac{1}{25}\right)$ Also, the circle passes through the point $A(2,0)\therefore$ radius of the circle

$=\sqrt{\left(2-\frac{2}{5}\right)^2+\left(0+\frac{1}{25}\right)^2}$

Thus, equation of the required circle is

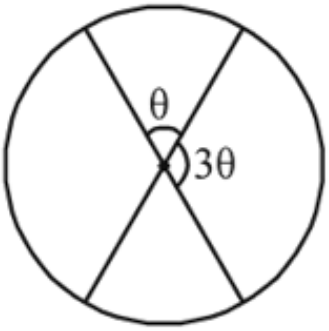
$=\left(x-\frac{2}{5}\right)^2+\left(y+\frac{1}{25}\right)^2=\frac{64}{25}+\frac{1}{625}$

or $25\left(x^2+y^2\right)-20x+2y-60=0$

Q.9 Answer:

$3a^2+2ab+3b^2=0$

Solution:



As per question area of one sector $=3$ area of another sector $\Rightarrow$ angle at centre by one sector $=3\times$ angle at centre by another sector Let one angle be θ then other $=3\theta$

Clearly $\theta+3\theta=180\Rightarrow\theta=45^\circ$

$\therefore$ Angle between the diameters represented by combined equation $ax^2+2(a+b)xy+by^2=0$ is 45°

$\therefore$ Using $\tan\theta=\frac{2\sqrt{h^2-ab}}{a+b}$

we get $\tan45^\circ=\frac{2\sqrt{(a+b)^2-ab}}{a+b}$

$\Rightarrow1=\frac{2\sqrt{a^2+b^2+ab}}{a+b}$

$\Rightarrow(a+b)^2=4\left(a^2+b^2+ab\right)$

$\Rightarrow a^2+b^2+2ab=4a^2+4b^2+4ab$

$\Rightarrow3a^2+3b^2+2ab=0$

@IITJEE_Advanced

Q.10 Answer:

$-r\sqrt{1+m^2} < c < r\sqrt{1+m^2}$

Solution:

Given line is $\mathrm{y} = \mathrm{mx} + \mathrm{c}$

and the given circle is $x^2 + y^2 = r^2$ **. . . Solving (1) and (2), we get**

$\left(1+m^2\right)x^2 + 2mcx + c^2 - r^2 = 0$

For two real distinct points of intersection, both the roots of (3) **must be real and distinct.**

$\therefore \quad 4m^2c^2 - 4\left(1+m^2\right)\left(c^2 - r^2\right) > 0$

$\Rightarrow \quad c^2 < r^2\left(1+m^2\right) \Rightarrow$

$-r\sqrt{1+m^2} < c < r\sqrt{1+m^2}$

Q.11 Answer:

$\alpha^2 + \beta^2 - a^2$

Solution:

Any point on the line at a distance r **from the point** $\mathrm{P}(\alpha, \beta)$ **is** $(\alpha + r\cos\theta, \beta + r\sin\theta)$

If this point lies on $\mathrm{x}^2 + \mathrm{y}^2 = \mathrm{a}^2$**, then** $\alpha^2 + r^2\cos^2\theta + 2\alpha r\cos\theta + \beta^2 + r^2\sin^2\theta + 2\beta r\sin\theta = a^2$

$\Rightarrow r^2 + 2r(\alpha\cos\theta + \beta\sin\theta) + \alpha^2 + \beta^2 = a^2$

$\Rightarrow r^2 + 2r(\alpha\cos\theta + \beta\sin\theta) + \alpha^2 + \beta^2 - a^2 = 0$

Now, if $PA = r_1$ **and** $PB = r_2$**, then** r_1 **and** r_2 **must be roots of this equation.**

$\therefore PA.PB = r_1 \cdot r_2 = \alpha^2 + \beta^2 - a^2$

Q.12 Answer:

$2\sqrt{\frac{r^2(a^2+b^2) - a^2b^2}{a^2+b^2}}$

Solution:

Length of chord $= 2\{$ **(radius)** $^2 - ($ **length of** $\perp$ **from centre of chord)** $^2\}^{1/2}$

$= 2\left\{r^2 - \left(\frac{-1}{\sqrt{(1/a^2)+(1/b^2)}}\right)^2\right\}^{1/2}$

$= 2\sqrt{\frac{r^2(a^2+b^2) - a^2b^2}{a^2+b^2}}$

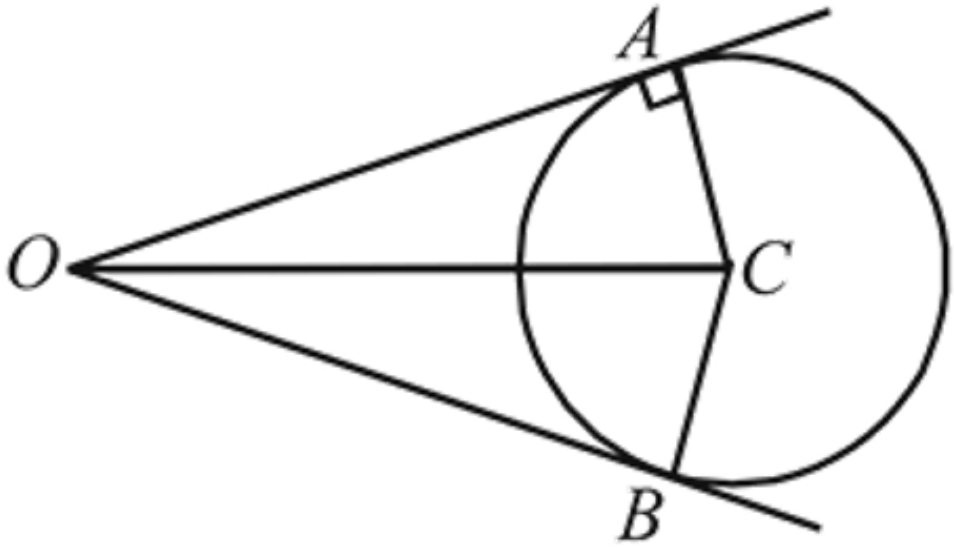
Q.13 Answer:

$\sqrt{c\left(g^2 + f^2 - c\right)}$

Solution:

@IITJEE_Advanced

(b) Area of quadrilateral $= 2[$ **area of** $\Delta OAC] = 2 \cdot \frac{1}{2} OA \cdot AC = \sqrt{S_1} \cdot \sqrt{g^2 + f^2 - c}$



Point is $(0, 0) \Rightarrow S_1 = c$

$\therefore$ **Area** $= \sqrt{c\,(g^2 + f^2 - c)}$

Q.14 Answer:

$2\mathrm{PN.AB}$

Solution:

Let the two given circles be $x^2 + y^2 + 2g_1 x + c = 0$

and $x^2 + y^2 + 2g_2 x + c = 0$

Their centres are $\mathrm{A}\,(-\mathrm{g}_1, 0)$ **and** $\mathrm{B}\,(-\mathrm{g}_2, 0)$

$\therefore \quad \mathrm{AB} = \mathrm{g}_1 - \mathrm{g}_2$

Let P **be the point** $(\mathrm{x}_1, \mathrm{y}_1)$ **. Then,** $\mathrm{PT} = \sqrt{\mathrm{x}_1^2 + \mathrm{y}_1^2 + 2\,\mathrm{g}_1 \mathrm{x}_1 + \mathrm{c}}$

$\mathrm{PT} = \sqrt{\mathrm{x}_1^2 + \mathrm{y}_1^2 + 2\,\mathrm{g}_2 \mathrm{x}_1 + \mathrm{c}}$

Radical axis of (1) **and** (2) **is** $2\,(\,\mathrm{g}_1 - \mathrm{g}_2)\,\mathrm{x} = 0$ **or** $\mathrm{x} = 0$

$\mathrm{PN} =$ **length of** $\perp$ **from** P **on radical axis** $= \mathrm{x}_1 \therefore \mathrm{PT}^2 - \mathrm{PT}^2$

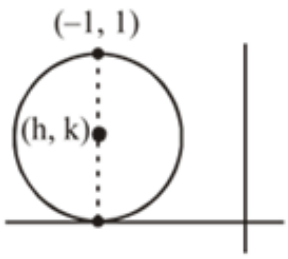
$= \left(x_1^2 + y_1^2 + 2g_1 x_1 + c\right) - \left(x_1^2 + y_1^2 + 2g_2 x_1 + c\right)$

$= 2\mathrm{x}_1\,(\,\mathrm{g}_1 - \mathrm{g}_2) = 2\mathrm{PN.AB}$

Q.15 Answer:

$k \geq \frac{1}{2}$

Solution:



@IITJEE_Advanced

The limiting case is when $(-1,1),(h,k)$ and point of contact are collinear
i.e., $2k = 1 \Rightarrow k = \frac{1}{2}$
$\therefore k \geq \frac{1}{2}$

Q.16 Answer:

$9x$ - $10y$ - $11 = 0$

Solution:

Let the center of the circle be $(-\mathrm{g}, -\mathrm{f})$.
Using the condition of orthogonality
$2\left(g_1 g_2 + f_1 f_2\right) = c_1 + c_2$
$2(2g - 3f) = 9 + c \ldots\ldots\ldots\ldots\ldots\ldots [1]$
$2\left(-\frac{5\,\mathrm{g}}{2} + 2\mathrm{f}\right) = -2 + \mathrm{c} \ldots\ldots\ldots\ldots\ldots [2]$
Subtract [2] from [1]:
$2\left(\frac{9g}{2} - 5f\right) = 11$
$\Longrightarrow 9\ \mathrm{g} - 10\mathrm{f} = 11$
Now for locus replace $-\mathrm{g}$ **by** x **and** $-\mathrm{f}$ **by** y **:**
Hence, $9\mathrm{x} - 10\mathrm{y} = 11$

Q.17 Answer:

$x^2 + y^2 - 4x - 6y + 5 = 0$

Solution:

Radius of the circle = perpendicular distance of $(2,3)$ **from** $x + y = 1$ **is** $\frac{4}{\sqrt{2}} = 2\sqrt{2}$ **The required equation will be** $(x-2)^2 + (y-3)^2 = 8 \Rightarrow x^2 + y^2 - 4x - 6y + 5 = 0$

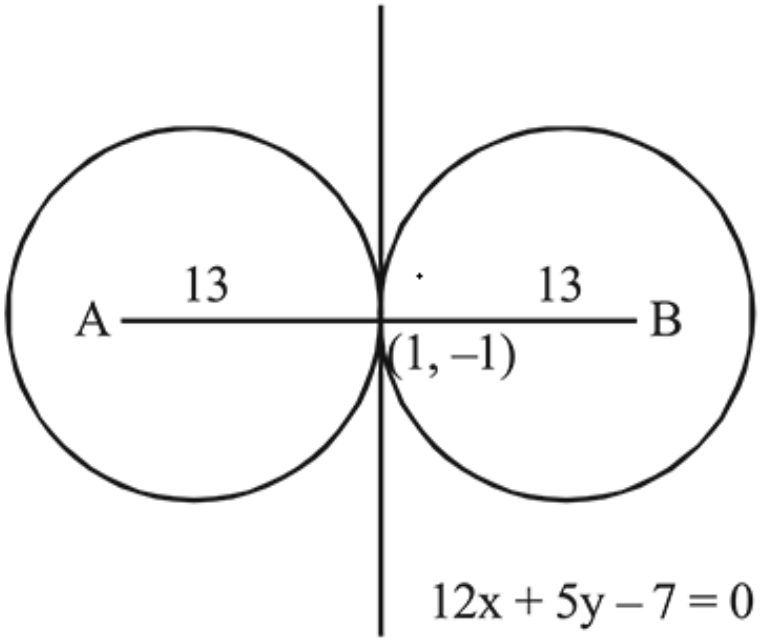
Q.18 Answer:

$(13,4),(-11,-6)$

Solution:

@IITJEE_Advanced

Let A, B, be the centres of the two circles, Slope of the common tangent $= -\frac{12}{5}$



$\therefore$ Slope of AB is $\tan\theta = -\frac{1}{-\frac{12}{5}} = \frac{5}{12}$

The point $(1, -1)$ lies on the line AB and the points A and B are at a distance 13 from the point $(1, -1)$ $\therefore$ Coordinates of A and B are

$(1 \pm 13\cos\theta, -1 \pm 13\sin\theta)$, where $\tan\theta = \frac{5}{12}$

i.e. $\left(1 \pm 13\frac{12}{13}, -1 \pm 13\frac{5}{13}\right)$ or $(1 \pm 12, -1 \pm 5)$

i.e. $(13, 4)$ and $(-11, -6)$

Q.19 Answer:

$x^2 + y^2 - 4x - 4y + 4 = 0$

Solution:

The equation of circle, which touches both the axes, is given by

$$x^2 + y^2 - 2rx - 2ry + r^2 = 0 \quad \ldots\ldots$$

Now, the centre (r, r) of this circle lies on the line

$$x + y = 4$$
$$r + r = 4 \Rightarrow r = 2$$

$\therefore$ **Put value of r in Eq. (i), we get**

$$x^2 + y^2 - 4x - 4y + 4 = 0$$

@IITJEE_Advanced

which is required equation of circle,

Q.20 Answer:

$1 : 2$

Solution:

Let $\mathrm{P}(\mathrm{h}, \mathrm{k})$ **be a point on the circle** $15x^2 + 15y^2 - 48x + 64y = 0$

Then the lengths of the tangents from $\mathrm{P}(\mathrm{h}, \mathrm{k})$ **to** $5x^2 + 5y^2 - 24x + 32y + 75 = 0$ **and** $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ **are**

$\mathrm{PT}_1 = \sqrt{\mathrm{h}^2 + \mathrm{k}^2 - \frac{24}{5}\,\mathrm{h} + \frac{32}{5}\mathrm{k} + 15}$

and $\mathrm{PT}_2 = \sqrt{\mathrm{h}^2 + \mathrm{k}^2 - \frac{48}{5}\,\mathrm{h} + \frac{64}{5}\mathrm{k} + 60}$

or $PT_1 = \sqrt{\frac{48}{15}h - \frac{64}{15}k - \frac{24}{5}h + \frac{32}{5}k + 15}$

$= \sqrt{\frac{32}{15}\mathrm{k} - \frac{24}{15}\,\mathrm{h} + 15}$

(Since (h, k) **lies on** $15\mathrm{x}^2 - 15\mathrm{y}^2 - 48\mathrm{x} + 64\mathrm{y} = 0$

$\therefore h^2 + k^2 - \frac{48}{15}h + \frac{64}{15}k = 0\big)$

and $\mathrm{PT}_2 = \sqrt{\frac{48}{15}\,\mathrm{h} - \frac{64}{15}\mathrm{k} - \frac{48}{5}\,\mathrm{h} + \frac{64}{5}\mathrm{k} + 60}$

$= \sqrt{-\frac{96}{15}\,\mathrm{h} + \frac{128}{15}\mathrm{k} + 60}$

$= 2\sqrt{-\frac{24}{15}\,\mathrm{h} + \frac{32}{15}\mathrm{k} + 15} = 2\mathrm{PT}_1$

$\Rightarrow \mathrm{PT}_1 : \mathrm{PT}_2 = 1 : 2$

@IITJEE_Advanced